Overview of Kalman Filter Theory and Navigation Applications
Day 3

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Day 3: Practical Kalman Filtering, Strapdown Navigation and Inertial Error Dynamics

Segments

- Practical Kalman Filtering: Some Numerical Considerations
- The Navigation Problem and Reference Frames
- Inertial Error Dynamics
- The Basic 9-State Kalman Filter
- Challenges of INS Filters

Some Practical Considerations (Numerical)

Topics

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- Kalman Filter Theory vs. Reality

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- Numerical Stability and Round-Off Errors

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- Numerical Stability and Round-Off Errors
- Matrix Inversion

Kalman Filter Theory vs. Reality

Kalman Assumption Recap

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- Kalman Assumptions

Kalman Filter Theory vs. Reality

Kalman Assumption Recap

- Kalman Assumptions
  - All states linear are modeled correctly
  - Noise parameters are accurately known
  - Initial values are appropriate
  - Process and measurement noise are both Gaussian: $w \sim \mathrm{N}(0, Q)$, and $v \sim \mathrm{N}(0, R)$
  - Process noise and measurement noise are uncorrelated
  - Measurements are taken at regularly scheduled intervals

Kalman Filter Theory vs. Reality

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- But continous filters unrealistic in practice: Always some sort of discretization needed
- Measurements almost never continuous
- Discrete filters acceptable provided continuous estimates are not required
- Suboptimality

Kalman Filter Theory vs. Reality

Sensor Selection

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  - The more the merrier, i.e., the more sensors the better
    Greater number of lower quality sensors can outperform smaller number of higher quality ones (see kalcompare.m)

Kalman Filter Theory vs. Reality

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  - The more the merrier, i.e., the more sensors the better Greater number of lower quality sensors can outperform smaller number of higher quality ones (see kalcompare.m)
  - Maximize the number of directly observable states

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Kalman Filter Theory vs. Reality

Sensor Selection

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  - Short-term / long-term complementarity (e.g, GPS vs. INS)

Kalman Filter Theory vs. Reality

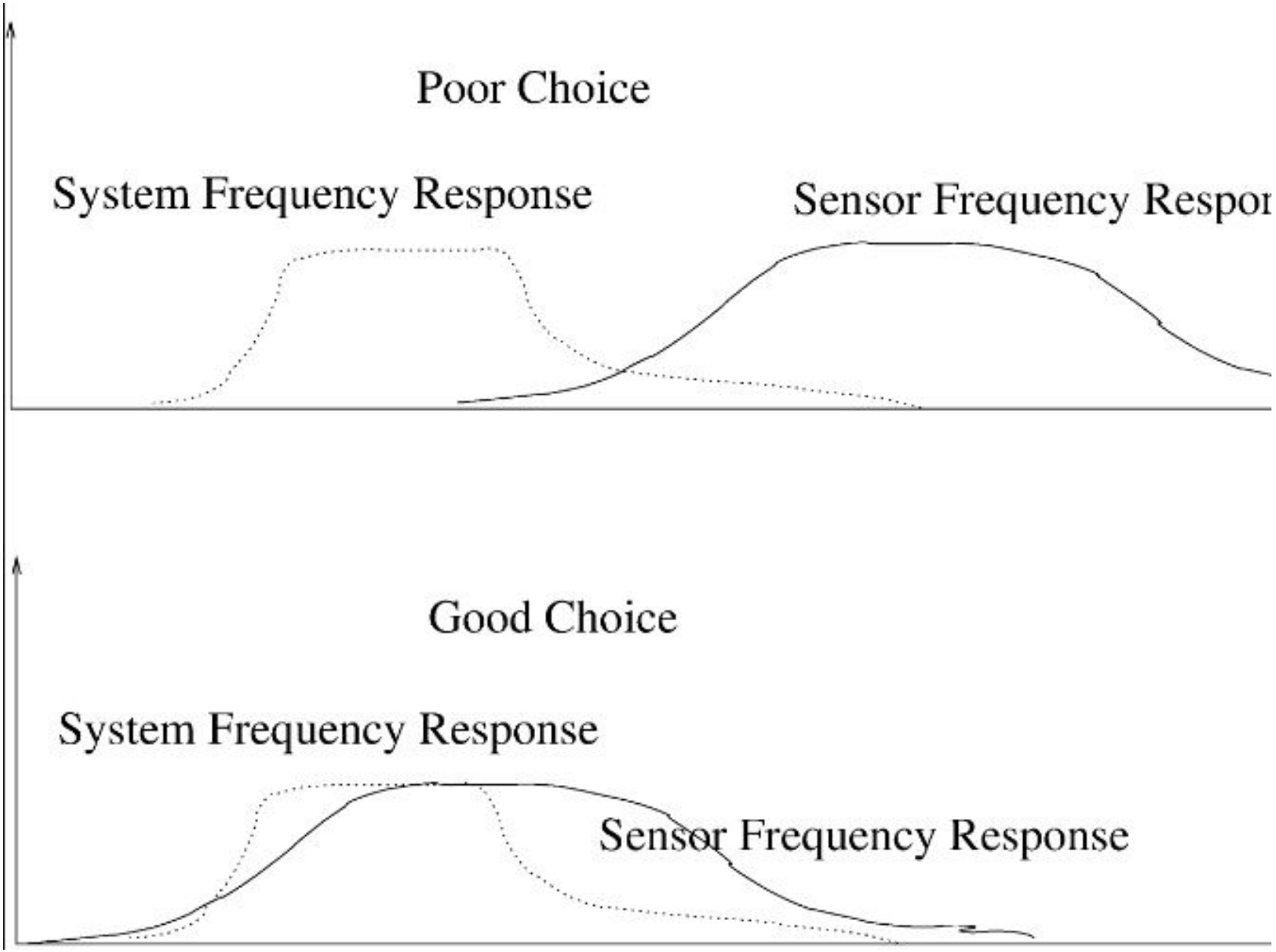
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  - Frequency response of sensor should match that of system

Kalman Filter Theory vs. Reality

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  - Short-term / long-term complementarity (e.g, GPS vs. INS)
  - Frequency response of sensor should match that of system
  - Noise spectrum of sensor should be maximally disjoint from that of system


Poor Choice
System Frequency Response
Sensor Frequency Respor
Good Choice
System Frequency Response
Sensor Frequency Response

Kalman Filter Theory vs. Reality

Kalman Filter Theory vs. Reality

- Choose sensors to match states.
  E.g., consider the state model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

  with the measurement model

$$z = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

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$$z = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

- Here x_1 is observable but x_2 is not, because there are no dynamic paths in $A = I$ flow infomation about x_1 into x_2

Kalman Filter Theory vs. Reality

Sensor Selection

- Examine the H matrix that you would use with this sensor!

Kalman Filter Theory vs. Reality

Sensor Selection

- Examine the H matrix that you would use with this sensor!
- If it renders some states unobservable, then consider adding additional sensors for those states, or replacing this sensor with one that is more favorable

Observability

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- Structural observability determined by Φ and H:
- If H is $1 \times n$, where n is the size of the state vector, then the state x_0 is said to be **observable** from n scalar measurements $\{z_0, \ldots, z_{n-1}\}$ if the block matrix

$$\begin{bmatrix} H^\top & \Phi^\top H^\top & \cdots & (\Phi^\top)^{n-1} H^\top \end{bmatrix}$$

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  is of rank n
- Will making measurements as governed by H actually enable us to estimate all of the states in the state vector?

Observability

Designing Observability into the Filter

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- Observability challenge for filters is often a matter of choosing right kinds of sensors
- Choosing the right measurement scheduling
- Certain kinds of divergence can be the result of poor observability
- Example: Let $H = [1 \;\; 0 \;\; 0 \;\; 0 \;\; 0]$, $\Phi = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$

Observability

Designing Observability into the Filter

- Observability challenge for filters is often a matter of choosing right kinds of sensors
- Choosing the right measurement scheduling
- Certain kinds of divergence can be the result of poor observability
- Example: Let $H = [1 \;\; 0 \;\; 0 \;\; 0 \;\; 0]$, $\Phi = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$
- What is the rank of $\begin{bmatrix} H^\top & \Phi H^\top & \Phi^2 H^\top & \Phi^3 H^\top & \Phi^4 H^\top \end{bmatrix}$?

Filter Divergence

Filter Divergence

- Kinds of Divergence

Filter Divergence

- Kinds of Divergence
- Causes and Remedies

Filter Divergence

Kinds of Divergence

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- Mean estimation error is supposed to converge to 0
  - $\tilde{x}^+(k) = \hat{x}^+(k) - x(k)$
  - $\mathrm{E}\left[\tilde{x}^+(k)\right] \to 0$
  - Estimate is unbiased: $\mathrm{E}\left[\hat{x}(k)\right] = x(k)$

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- Covariance matrix for estimation error is same as state estimation error covariance matrix:
  $P^+(k) = \mathrm{E}\left[\tilde{x}^+(k)\left[\tilde{x}^+(k)\right]^\top\right]$

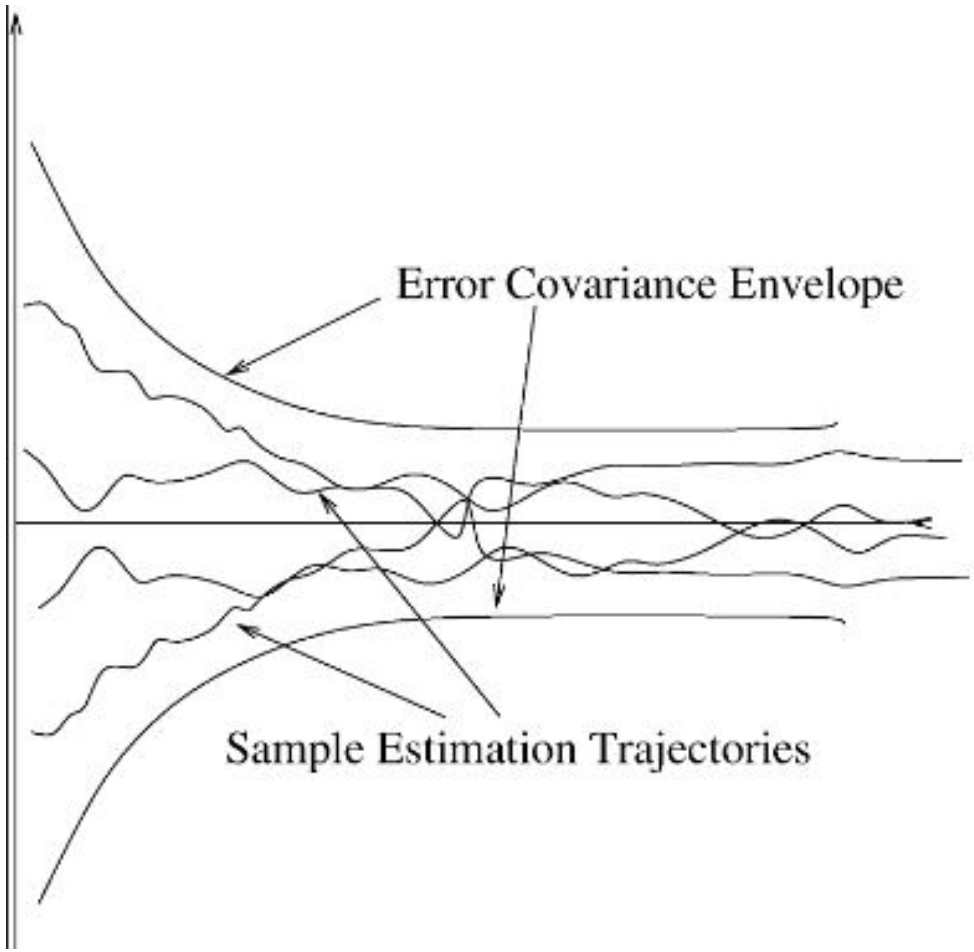
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- For optimal filter, P will indicate convergence or divergence.
  - One common form of divergence due to round-off will send P to zero
- For suboptimal filters, P will not necessarily indicate divergence
  - P may seem to indicate nominal steady-state values and yet the state estimate may go diverge from the true value


Error Covariance Envelope
Sample Estimation Trajectories

Kinds of Divergence

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- Going to sleep: error covariance becomes so small that measurements are no longer believed by the filter
- Loss of symmetry in P
- Loss of positive definiteness: negative numbers on the diagonal!

Numerical Stability

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- Numerical stability: small errors at each computation step do not grow without bound
- Matrix Conditioning (will address in discussing SRIF)

Round-Off Errors

Joseph Form

- Computer round-off in floating point arithmetic

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Round-Off Errors

Joseph Form

- Computer round-off in floating point arithmetic
- Unit round-off value is largest number ε such that $1 + \varepsilon = 1$
- In Matlab, the global symbol "eps" stores the unit round-off value
- Round-off errors generally plague the classical covariance update algorithm

$$P^{+}(k) = [I - K(k)H(k)]\, P^{-}(k)$$

- Alternative, called the Joseph form, is more robust:

$$P^{+}(k) = [I - K(k)H(k)]\, P^{-}(k)\, [I - K(k)H(k)]^{\top} + K(k)R(k)K(k)^{\top}$$

Round-Off Errors

De Vries Mechanization

- This form in turn leads to the De Vries mechanization of the Kalman gain equation. To see this, first expand the Joseph form (omitting subscripts for clarity):

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$$P(+) = P(-) - KHP(-) - P(-)H^\top K^\top + KHP(-)H^\top K^\top + KRK^\top$$

Round-Off

De Vries

- Define

$$F = PH^{\top} \tag{1}$$

and

$$D = HF + R. \tag{2}$$

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$$D = HF + R. \tag{2}$$

- We note that D is symmetric, because

$$\begin{aligned} D^\top &= [HF + R]^\top \\ &= [HPH^\top + R]^\top \\ &= [HPH^\top]^\top + R^\top \\ &= [H^\top]^\top P^\top H^\top + R^\top \\ &= HPH^\top + R \qquad \text{(because } P \text{ and } R \text{ are symmetric)} \\ &= HF + R \\ &= D. \end{aligned}$$

- The gain in standard Kalman gain equation

$$K(k) = P^{-}(k)\, H(k)^{\top} \left[H(k) P^{-}(k)\, H(k)^{\top} + R(k) \right]^{-1}$$

can be written as

$$K = FD^{-1}.$$

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- Using Eqs. (1) and (2), we compute

$$\begin{aligned} P(+) &= P(-) - KF^{\top} - FK^{\top} + KDK^{\top} \\ &= P(-) - K \left[F^{\top} - \frac{DK^{\top}}{2} \right] - \left[F - \frac{KD}{2} \right] K^{\top}. \end{aligned}$$

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- Letting $G = F - \frac{KD}{2}$, we obtain

$$P(+) = P(-) - GK^{\top} - \left[GK^{\top}\right]^{\top}. \tag{3}$$

- The De Vries mechanization implements Eq. (3) by subtracting the ij^{th} element of $GK^{\top}$ from both the ij^{th} and ji^{th} elements of $P(-)$.

- The De Vries mechanization implements Eq. (3) by subtracting the ij^{th} element of $GK^\top$ from both the ij^{th} and ji^{th} elements of $P(-)$.
- The De Vries form of the covariance update equation is valid for any arbitrary gain matrix K, not just the optimal gains computed from Eqs. (1) and (2).

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Numerical Stability

Matrix Inversion

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Matrix Inversion

- Matrix inversion is always risky due to potential ill-conditioning
- One way to avoid this is to process measurements sequentially as scalars
- Or at least as low-dimensional matrices

Exercises

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- 1. Let $v = [1 \quad -2]^\top$. Compute the positive definite, symmetrix matrix $M = vv^\top$
- 2. Use the following algorithm (Grewal & Andrews) to factor M into $UDU^\top$ form:

Exercises

```
for j=m:-1:1
    for i=j:-1:1
        covar=P(i,j);
        for k=j+1:m
            covar=covar-U(i,k)*D(k,k)*U(j,k);
        end
        if i==j
            D(j,j)=covar;
            U(j,j)=1;
        else
            U(i,j)=covar/D(j,j);
        end
    end
end
```

Segment 10
The Navigation Problem and Reference Frames
Topics

- Basic Navigation Problem
- Earth Referenced Navigation
- Coordinate Frame Conversion
- Basic Strapdown Navigation Equations

Basic Navigation Problem

- Where am I?
- Where am I headed?
- How fast am I going?

Basic Navigation Problem

- What's my position?
- What's my heading and attitude?
- What's my velocity?

Solution in Inertial Space

- In inertial space, it's no problem!
- Just integrate the acceleration once to get velocity, twice to get position.

$$v(T) = v(0) + \int_0^T \frac{d^2 r}{dt^2} dt$$

$$r(T) = r(0) + \int_0^T v(T) dt$$

- And integrate the angular rate once to get attitude.

Solution in Inertial Space

- Solution easy in inertial space because ...
  - inertial instruments sense acceleration and rotational rate with respect to inertial coordinate frame
  - inertial frame attached to sensor platform
  - Note: You still need to rotate accelerations from body to inertial coordinates prior to integration (strapdown)
- But we don't necessarily want the answer to our questions expressed in terms of inertial coordinates!

Earth Referenced Navigation

- Terrestial navigation seeks to answer nav questions with respect to earth
  - Global earth reference
  - Local or relative navigation
- Complications:
  - Earth surface is curved (ellipsoid)
  - Earth is turning (can usually ignore motion about the Sun for terrestrial nav)

Coordinate Frames

- Inertial
- Earth Fixed
- Local Level
- Body

Coordinate Frames

- Technically speaking, all frames are earth-centered
- Practically speaking, we think of the Body frame as centered in the body of the vehicle (usually at center of gravity).
- Local Level is thought to be located at surface of earth directly beneath the vehicle.

Inertial Reference Frames

- Non-Rotating
- Non-Accelerating
- Fixed or moving uniformly with respect to another inertial frame (fixed stars)

Inertial Reference Frames

- Nav problem "easy" in inertial frames
- Integrate gyro rates once to get attitude (rotational motion is absolute)
- Rotate accelerometer outputs to inertial frame using attitude matrix, then integrate to get inertial velocity; integrate velocity to get position.
- Would still have to contend with instrument imperfections.

Inertial Reference Frame Integrations

$$\ddot{r}(t) = a_{\text{sf}} + \text{other external forces} \tag{4}$$

$$v(T) = \dot{r}(T) = v(0) + \int_0^T a_{\text{sf}} dt + \int_0^T (\text{other external forces}) dt \tag{5}$$

$$r(T) = r(0) + \int_0^T v dt \tag{6}$$

Earth Fixed = ECEF = Earth Centered Earth Fixed

- Rotating frame, i.e., non-inertial
- Cartesian (X, Y, Z in meters)
- Spherical (latitude ϕ, longitude λ, altitude
  - geodetic

Local Level

- Tangent to earth's surface but translated to center of earth.
- Vertical Z-axis is normal to earth's surface
- X, Y-axes are level
- Several kinds: Y north-slaved, wander (0 transport rate) or free azimuth
- Y makes wander angle α with respect to true north (ENU=East, North, Up).

Body Frame

- Attached to vehicle
- For aircraft:
  - X out the nose
  - Y out the right wing
  - Z down

Coordinate Frame Conversion

- All frames are earth-centered, hence differ only by rotation
- Rotation represented by direction cosine matrices (DCM)
  - Represents the angles made by each of the basis vectors of one frame with those of the other
- C_A^B = DCM transformation from frame A to frame B

Coordinate Frame Conversion

- "Direction Cosine" comes from dot product:

$$\langle x, y \rangle = x \cdot y = \|x\| \, \|y\| \cos \theta_{xy}$$

  where θ_{xy} is the angle between the two vectors x and y.
- If x and y are unit vectors, then $\|x\| = \|y\| = 1$ and we have:

$$\langle x, y \rangle = x \cdot y = \cos \theta_{xy}$$

Coordinate Frame Conversion

- If $U = \{u_1, u_2, u_3\}$ and $V = \{v_1, v_2, v_3\}$ are two sets of orthonormal basis vectors, then the DCM relating one frame to the other is given by:

$$C_U^V = \begin{bmatrix} \langle u_1, v_1 \rangle & \langle u_1, v_2 \rangle & \langle u_1, v_3 \rangle \\ \langle u_2, v_1 \rangle & \langle u_2, v_1 \rangle & \langle u_2, v_3 \rangle \\ \langle u_3, v_1 \rangle & \langle u_3, v_2 \rangle & \langle u_3, v_3 \rangle \end{bmatrix}$$
$$= \begin{bmatrix} \cos\theta_{11} & \cos\theta_{12} & \cos\theta_{13} \\ \cos\theta_{21} & \cos\theta_{22} & \cos\theta_{23} \\ \cos\theta_{31} & \cos\theta_{32} & \cos\theta_{33} \end{bmatrix}$$

Coordinate Frame Conversion

- DCMs are orthogonal matrices $\Rightarrow \left[C_A^B\right]^{-1} = \left[C_A^B\right]^\top$
- Third row (or column) is vector cross product of other two $\Rightarrow$ only six independent parameters
- DCMs satisfy: $C_B^A C_C^B = C_C^A$

Coordinate Frame Conversion: Most important DCMs

- Body to Local Level: C_B^L
  - sometimes called Body to Nav and notated C_B^N
  - yields attitude info (Euler angles for roll, pitch, yaw)
- Local Level to Earth: C_L^E
  - yields position info and also used to transform earth rate to local level

Coordinate Frame Conversion: Most important DCMs

- Visualize C_L^E's determination of position as follows:
  - As vehicle moves, local level frame changes its orienation with respect to earth's center in order to maintain local level with respect to earth's surface
  - If you think of local level frame as attached to earth's surface, you get distracted by the displacement of the frame
  - If you move local level frame to center of earth, the displacement disappears and all that remains are the rotations

Strapdown Navigation: Strapdown vs. Gimbaled

- Gimbaled INS:
  - Gyro stabilized platform
  - Gimbal assembly and torque motors
  - Resolvers, slip rings, bearings, etc. to control platform
  - Goal: Maintain stable, locally level or inertial reference frame

Strapdown Navigation: Strapdown vs. Gimbaled

- Gimbaled INS Issues
  - Reliability and Cost: Lots of moving parts!
  - Body rate and acceleration information not directly available
  - Rate signals are generally noisy and have low bandwidth

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS:
  - Inertial instruments strapped to body frame
  - Requires local level accelerations to be computed using direction cosine matrix C_B^L
  - C_B^L derived from gyro-sensed body rates
  - Strapdown systems can be implemented with either rate gyros or attitude gyros

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS Advantages
  - Immediate detection of body rates and accelerations
  - Fewer moving parts
  - Potential redundancy advantages
- Gimbaled systems are nearly obsolete

Basic Strapdown Navigation Equations

- Need to compute attitude, velocity, and position
- I.e., need to compute C_B^L and C_L^E
- These DCMs are dynamic quantities
- C_B^L determines attitude and can be rapidly varying
- C_L^E determines position and is slowly varying (relative to attitude dynamics)

Rotating Coordinate Frames

- We first need to understand how an arbitrary vector and its time derivatives in one coordinate frame are expressed in another frame that is rotating with respect to the first.
- Let A and B denote two coordinate frames and assume that B is rotating with respect to A.
- We use the notation ω_{AB}^{A} to express the direction and magnitude of the rotation vector as seen in the A frame.

Rotating Coordinate Frames

- Every rotation vector can also be expressed as a skew symmetric matrix.
- If ω is a rotation vector (without mentioning the frames involved), then

$$\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}.$$

Rotating Coordinate Frames

- Since multiplication of a vector by the skew symmetric matrix equivalent is the same thing as left multiplying the cross product of the original vector, we can write:

$$\Omega v = \omega \times v$$

- Therefore the notation $[\ \omega\times\ ] = \Omega$ is common[1]

[1] One must be careful in making this identification since left matrix multiplication is associative while left cross product multiplication is not.

Rotating Coordinate Frames

- Let v be an arbitrary vector in 3-space.
- Let $\left\{e_A^1, e_A^2, e_A^3\right\}$ be orthonormal basis vectors in the A frame
- Let $\left\{e_B^1, e_B^2, e_B^3\right\}$ be orthonormal basis vectors in the B frame[2].

[2]We reverse the roles of superscripts and subscripts for basis vectors so that we may employ the Einstein summation convention on occasion.

Rotating Coordinate Frames

- Thus, we can write v as $\sum_{i=1}^{3} v_i^A e_A^i$ and also $\sum_{i=1}^{3} v_i^B e_B^i$.
- This is one and the same vector expressed in two different coordinate frames[3].

[3] A vector is a linear combination of basis vectors and not necessarily an ordered triple of real numbers.

Rotating Coordinate Frames

- Note that, with respect to an arbitrary inertial reference frame, we have for the time derivative:

$$v = \sum_{i=1}^{3} v_i^A e_A^i = \sum_{i=1}^{3} v_i^B e_B^i$$

$$\Rightarrow \dot{v} = \frac{d}{dt}\left[\sum_{i=1}^{3} v_i^A e_A^i\right] = \frac{d}{dt}\left[\sum_{i=1}^{3} v_i^B e_B^i\right]$$

$$\Rightarrow \sum_{i=1}^{3}\left(\dot{v}_i^A e_A^i + v_i^A \dot{e}_A^i\right) = \sum_{i=1}^{3}\left(\dot{v}_i^B e_B^i + v_i^B \dot{e}_B^i\right)$$

Rotating Coordinate Frames

- In the special case that their respective coordinate frames are not rotating relative to one another, each $\dot{e}_A^i = 0 = \dot{e}_B^i$.
- Nevertheless, regardless of rotation, we must always have:

$$\sum_{i=1}^{3} \dot{v}_i^A e_A^i = \sum_{i=1}^{3} \dot{v}_i^B e_B^i. \tag{7}$$

(Note that, in general, $\dot{v}_i^A \neq \dot{v}_i^B$.)

Rotating Coordinate Frames

- Let frame B rotate with respect to frame A at a constant rotational rate ω_{AB}^{A} as expressed in A frame coordinates.
- Since B is not rotating with respect to itself, let us agree to label the quantity in Eq. (7) as $\dot{v}^{B}$.
- We'll assume that A is an inertial frame and thus $\dot{v} = \dot{v}^{A}$.

Rotating Coordinate Frames

- When viewed from the A frame, however, it is not true that each of the $\dot{e_i}^B$ is zero, since these unit vectors are slaved to the B frame and B is rotating with respect to A at rate ω_{AB}^A.
- Therefore, we have:

$$\begin{aligned}
\dot{v}^A &= \sum_{i=1}^{3} \dot{v}_i^A e_A^i + \sum_{i=1}^{3} v_i^B \dot{e}_B^i \\
&= \sum_{i=1}^{3} \dot{v}_i^B e_B^i + \sum_{i=1}^{3} v_i^B \dot{e}_B^i \quad \text{(by Eq. (7))} \qquad (8) \\
&= \dot{v}^B + \sum_{i=1}^{3} v_i^B \dot{e}_B^i.
\end{aligned}$$

Rotating Coordinate Frames

- The last sum on the right side of Eq. (8) can be rewritten as the cross product $\omega_{AB}^{A} \times v$, because $\dot{e}_B^i = \omega_{AB}^{A} \times e_B^i$, and

$$\begin{aligned}\sum_{i=1}^{3} v_i^B \dot{e}_B^i &= \sum_{i=1}^{3} v_i^B \omega_{AB}^{A} \times e_B^i \\ &= \omega_{AB}^{A} \times \sum_{i=1}^{3} v_i^B e_B^i \\ &= \omega_{AB}^{A} \times v.\end{aligned}$$

Rotating Coordinate Frames

- We thus arrive at the all-important Coriolis Equation which is used often in navigation analysis:

$$\dot{v}^A = \dot{v}^B + \omega_{AB}^A \times v^B. \tag{9}$$

Alternatively, we can express Eq. (9) in skew-symmetric matrix form using $\Omega_{AB} = [\,\omega_{AB}\times\,]$:

$$\dot{v}^A = \dot{v}^B + \Omega^A_{AB} v^B. \tag{10}$$

Note that we must use superscripts on both Ω_{AB} and v as soon as we make a commitment to a particular coordinate frame representation.[4].

[4]Since matrices are defined in terms of their action on basis vectors, it is appropriate to specify the coordinate system. On the other hand, if Ω_{AB} is taken to be a linear transformation, then superscripts are not necessary.

Strapdown Navigation Equations

- Three differential equations
  - Equation for body to local level transformation C_B^L
  - Equation for velocity (uses C_B^L)
  - Equation for local level to earth for position DCM C_L^E

Strapdown Navigation Equations

- Two Algebraic Equations
  - Equation for converting velocity to transport rate (to be explained)
  - Equation for converting earth rate to local level coordinates

Strapdown Navigation Equations
The Three Differential Equations

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L \tag{11}$$

$$\dot{v}^L = C_B^L a_{\mathsf{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L \tag{12}$$

$$\dot{C}_L^E = C_L^E \Omega_{EL}^L \tag{13}$$

Strapdown Navigation Equations
The Two Algebraic Equations

$$\omega_{EL}^{L} = \frac{1}{r} u_r^L \times v^L \text{ (wander azimuth } \rho_z = 0) \tag{14}$$

$$\omega_{IE}^{L} = \left[C_L^E \right]^\top \omega_{IE}^{E} \tag{15}$$

Strapdown Navigation Equations
The C_B^L Equation (11)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- All the Ω matrices represent rotation vector cross products in skew-symmetric matrix format
  - Ω_{IB}^B is rotational rate of aircraft in body coordinates
  - Ω_{IE}^L is earth rate in local level coordinates
  - Ω_{EL}^L is transport rate in local level coordinates

Strapdown Navigation Equations: Detour into Cross Product Matrices

- The cross product of $\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$ with any other vector v is the same as multiplying v on the left by $\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$
- $\omega \times v = \Omega v$

Strapdown Navigation Equations
The C_B^L Equation (11)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- Transport rate is the rotational rate of the aircraft due to motion over the surface of the earth. Is a function of velocity.
- Ω_{IB}^B is total inertial angular rate sensed by gyros and includes effects of transport and earth rates.
- Hence transport and earth rate effects have to be subtracted off.

Strapdown Navigation Equations
The Strapdown Velocity Equation (12)

$$\dot{v}^L = C_B^L a_{\text{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L$$

- Sometimes referred to as the velocity rate equation
- a_{sf}^B is specific force sensed by accelerometers in body coordinates (because the accels are strapped to the body – hence the term strapdown)

Strapdown Navigation Equations
The Transport Rate Equation (13)

$$\dot{C}_L^E = C_L^E \Omega_{EL}^L$$

- Connected to velocity rate equation by the first algebraic equation (14):

$$\omega_{EL}^L = \frac{1}{r} u_r^L \times v^L$$

- Eq. (13) Sometimes referred to as position rate equation, because C_L^E directly implies the vehicle position

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Geographic)

- North transport rate:

$$\rho_N = \frac{v_E}{R_e}\left[1 - \frac{h}{R_e} - e\sin^2\phi\right] \tag{16}$$

- East transport rate:

$$\rho_E = -\frac{v_N}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3\cos^2\phi\right)\right] \tag{17}$$

where e is ellipticity (approx. $\frac{1}{298}$) and R_e is mean equatorial earth radius

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Geographic)

- Latitude Rate:

$$\dot{\phi} = -\rho_E \tag{18}$$

- Longitude Rate:

$$\dot{\lambda} = \rho_N \sec \phi \tag{19}$$

Strapdown Navigation Equations

- Geographic (north-slaved) mechanization has singularities at poles
- Two other approaches to local level coordinates:
  - Free azimuth: vertical component of transport rate $\rho_z =$ additive inverse of vertical component of earth rate
  - Wander Azimuth: vertical component of transport rate $\rho_z = 0$

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- X Transport Rate:

$$\rho_x = -\frac{v_y}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{22}^2 - d_{21}^2\right)\right] - \frac{v_x}{R_e}\left(2d_{21}d_{22}\right)e \tag{20}$$

- Y Transport Rate:

$$\rho_y = \frac{v_x}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{21}^2 - d_{22}^2\right)\right] + \frac{v_y}{R_e}\left(2d_{21}d_{22}\right)e \tag{21}$$

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- Azimuth Rate:

$$\rho_z = \begin{cases} -\Omega_e d_{23}, & \text{if free azimuth;} \\ 0, & \text{if wander azimuth} \end{cases}$$

- In equations above, the matrix $[d_{ij}]$ is the direction cosine matrix C_L^E from local level (L) frame to earth centered earth fixed (E) frame.

Exercises

1. Compute direction cosine matrix C_U^V, where

$$U = \{(1,0,0),(0,1,0),(0,0,1)\}$$

and

$$V = \{(-1,0,0),(0,1,0),(0,0,-1)\}$$

2. Suppose you are starting at the equator and travelling due north, and your velocity is 5 m/sec. Compute the transport rates ρ_n and ρ_e associated with this velocity and heading. You may use the following constants: $\Omega_e = 7.2921151467 \times 10^{-5}$ rad/sec and $R_e = 6,378,137$ meters

Inertial Error Dynamics

Topics

Inertial Error Dynamics

Topics

- Imperfect Inertial Instruments

Inertial Error Dynamics

Topics

- Imperfect Inertial Instruments
- From Strapdown Equations to Error Equations

Inertial Error Dynamics

Topics

- Imperfect Inertial Instruments
- From Strapdown Equations to Error Equations
- Initialization: Alignment

Inertial Error Dynamics

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections

Inertial Error Dynamics

Imperfect Inertial Instruments

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- Bias, scale factor, g^2 terms, noise, random walks, etc.

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Imperfect Inertial Instruments

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- Bias, scale factor, g^2 terms, noise, random walks, etc.
- Inertial sensor clusters also have errors: misalignments
- Many error sources are calibrated out using rate tables during ATP
- Residual effects will influence strapdown nav equations

Inertial Error Dynamics

Imperfect Inertial Instruments

- Strapdown nav equations very sensitive to gyro errors

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- Gyro biases yield erroneous C_B^L matrix which in turn causes erroneous accelerations to be computed in local level frame

- Strapdown nav equations very sensitive to gyro errors
- Gyro biases yield erroneous C_B^L matrix which in turn causes erroneous accelerations to be computed in local level frame
- Thus, gyro errors affect not only attitude, but also position and velocity

- Gyros

$$\epsilon\omega_x = \Delta B_{gx} + \Delta K_{gx}\omega_x + \theta_{gxy}\omega_y + \theta_{gxz}\omega_z$$

$$\epsilon\omega_y = \Delta B_{gy} + \Delta K_{gy}\omega_y + \theta_{gyz}\omega_z + \theta_{gyx}\omega_x$$

$$\epsilon\omega_z = \Delta B_{gz} + \Delta K_{gz}\omega_z + \theta_{gzx}\omega_x + \theta_{gzy}\omega_y$$

where ΔB_{gi} is the ith gyro bias, ΔK_{gi} the ith gyro scale factor, and θ_{gij} is the misalignment of the ith gyro axis toward the jth gyro axis.

- Accelerometers

$$\epsilon a_x = \Delta B_{ax} + \Delta K_{ax} a_x + \theta_{axy} a_y + \theta_{axz} a_z$$

$$\epsilon a_y = \Delta B_{ay} + \Delta K_{ay} a_y + \theta_{ayz} a_z + \theta_{ayx} a_x$$

$$\epsilon a_z = \Delta B_{az} + \Delta K_{az} a_z + \theta_{azx} a_x + \theta_{azy} a_y$$

where ΔB_{ai} is the ith accelerometer bias, ΔK_{ai} the ith accelerometer scale factor, and θ_{aij} is the misalignment of the ith accelerometer axis toward the jth accelerometer axis.

- Bias in both types of instruments can be further broken down into several components:
  - fixed bias – constant from turn-on to turn-on – should be calibrated out
  - bias stability – varies from turn-on to turn-on – can't be calibrated out, but can be characterized in terms of variance – can model as random constant
  - bias drift – usually modeled as random walk

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- Scale factor can also be modeled as a sum of three components
- Often in nav filters, scale factor residual is modeled as a random constant
- Important to keep in mind that the contribution of scale factor error is a function of input (rate or acceleration)

- In expanding the output error as a Taylor series function of input, bias, scale factor, and misalignment show up as zero and first order terms.

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- In expanding the output error as a Taylor series function of input, bias, scale factor, and misalignment show up as zero and first order terms.
- Higher order terms are also present, especially second order terms.
- In some calibration and alignment filters, such higher order effects are modelled
- SNU-84-1 includes error budgets associated with all the combined, higher order effects.

$$
\begin{aligned}
f(x,y) = f(x_0,y_0) &+ f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) \\
&+ \frac{1}{2!}(f_{xx}(x_0,y_0)(x-x_0)^2 \\
&+ f_{xy}(x_0,y_0)(x-x_0)(y-y_0) \\
&+ f_{yy}(x_0,y_0)(y-y_0)^2) + \cdots
\end{aligned}
$$

Taylor Series provides basis for all error analysis.

Inertial Error Dynamics

From Strapdown Equations to Error Equations

Inertial Error Dynamics

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments

Inertial Error Dynamics

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Inertial Error Dynamics

From Strapdown Equations to Error Equations

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- Inertial instrument error sources (bias, scale factor errors, etc.) generate erroneous rate and acceleration inputs to the nav equations

Inertial Error Dynamics

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments
- Instruments are not perfect
- Inertial instrument error sources (bias, scale factor errors, etc.) generate erroneous rate and acceleration inputs to the nav equations
- Need to figure out the effects on the outputs of these equations

- Nature of Errors

Inertial Error Dynamics

Strapdown Error Equations

- Nature of Errors
- Error Operator

Inertial Error Dynamics

Strapdown Error Equations

- Nature of Errors
- Error Operator
- Error Equations

- Errors propagate according to derived equations

Inertial Error Dynamics

Strapdown Error Equations: Nature of Errors

- Errors propagate according to derived equations
- Errors are generally small, so approximating assumptions help simplify the math

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Strapdown Error Equations: Nature of Errors

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  - linearize otherwise non-linear dynamics

- Errors propagate according to derived equations
- Errors are generally small, so approximating assumptions help simplify the math
  - linearize otherwise non-linear dynamics
  - small angle assumptions: $\sin\theta \approx \theta$

- If A is any quantity, we define error operator as follows:

$$\delta A = \hat{A} - A, \tag{22}$$

where A is the true quantity of interest and $\hat{A}$ is an apparent quantity. The latter could be a measured value, and estimated value, or some computed value. This means that

$$\hat{A} = A + \delta A$$

i.e., the apparent value is the true value plus some error.

- Error operator satisfies:

$$\delta(AB) = \delta A \cdot B + A \cdot \delta B \tag{23}$$

- Error operator satisfies:

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- Proof:

$$\begin{aligned}
\delta(AB) &= \widehat{AB} - AB \\
&= \hat{A}\hat{B} - AB \\
&= (A + \delta A)(B + \delta B) - AB \\
&= AB + A \cdot \delta B + \delta A \cdot B + \delta A \cdot \delta B - AB \\
&= \delta A \cdot B + A \cdot \delta B
\end{aligned}$$

since we may assume that $\delta A \cdot \delta B \approx 0$.

- Applying these to the three differential equations, yields three error equations:

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L \tag{24}$$

$$\delta \dot{v}^L = \delta C_B^L a_{sf}^B + C_B^L \delta a_{sf}^B + \delta g' - \left(2\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) v^L - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) \delta v \tag{25}$$

$$\delta \dot{C}_L^E = \delta C_L^E \Omega_{EL}^L + C_L^E \delta \Omega_{EL}^L \tag{26}$$

- Introducing **small angular rotations** for tilt and position

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  - $\widehat{C}_B^L$, the computed (i.e., apparent) version of C_B^L is related additively to the true version by the error δC_B^L

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  - $\widehat{C}_B^L$, the computed (i.e., apparent) version of C_B^L is related additively to the true version by the error δC_B^L

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  - It is also related to C_B^L by a small direction cosine matrix relating the local level frame to the computer frame $\widehat{L}$:

$$\widehat{C}_B^L = C_{\widehat{L}}^L C_B^L$$

- Substituting the second equation into the first allows us to solve for δC_B^L:

$$\begin{aligned}\delta C_B^L &= \widehat{C}_B^L - C_B^L \\ &= \widehat{C_L^L} C_B^L - C_B^L \\ &= (\widehat{C_L^L} - I) C_B^L\end{aligned}$$

Now, let's introduce $\Phi = I - \widehat{C_L^L}$ to obtain:

$$\delta C_B^L = -\Phi C_B^L$$

- Using small angle approximations, $C_L^{\widehat{L}}$ can be written in the form

$$C_L^{\widehat{L}} = \begin{bmatrix} 1 & \phi_z & -\phi_y \\ -\phi_z & 1 & \phi_x \\ \phi_y & -\phi_x & 1 \end{bmatrix}.$$

Thus, $\Phi = I - C_L^{\widehat{L}}$ is seen to be a skew-symmetric rotation matrix:

$$\Phi = \begin{bmatrix} 0 & -\phi_z & \phi_y \\ \phi_z & 0 & -\phi_x \\ -\phi_y & \phi_x & 0 \end{bmatrix}.$$

- Φ is therefore equal to the cross product operator associated with a vector $\phi^L = \begin{bmatrix} \phi_x^L & \phi_y^L & \psi_z^L \end{bmatrix}^\top$. I.e.,

$$\Phi = \left[\phi^L \times \right]$$

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- The three components of ϕ^L will become three states in our inertial error dynamics Kalman filter.

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$$\Phi = \left[\phi^L \times \right]$$

- The three components of ϕ^L will become three states in our inertial error dynamics Kalman filter.
- ϕ_x and ϕ_y are called tilts, while ψ_z is called the azimuth error.

- By a similar derivation, we can introduce a small angular position angle error $e^L = \begin{bmatrix} e_x^L & e_y^L & e_z^L \end{bmatrix}^\top$

- By a similar derivation, we can introduce a small angular position angle error $e^L = \begin{bmatrix} e_x^L & e_y^L & e_z^L \end{bmatrix}^\top$
- e^L is associated with δC_E^L just as ϕ^L is associated with δC_B^L:

$$\delta C_E^L = -E^L C_E^L,$$

where E^L is the skew-symmetric matrix form of e^L.

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- e^L is associated with δC_E^L just as ϕ^L is associated with δC_B^L:

$$\delta C_E^L = -E^L C_E^L,$$

  where E^L is the skew-symmetric matrix form of e^L.
- The three components of e^L yield the three position error states in a typical navigation Kalman filter.

- We are working to simplify Eq. (24):

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L \right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L \right) \delta C_B^L$$

- We are working to simplify Eq. (24):

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L$$

- We have an expression for δC_B^L.

- Consider the earth rate error term $\delta\Omega^L_{IE}$. Since $\Omega^L_{IE} = C^L_E \Omega^E_{IE}$ and $\delta\Omega^E_{IE} = 0$, we derive:

$$\delta\Omega^L_{IE} = \delta C^L_E \Omega^E_{IE}.$$

- Consider the earth rate error term $\delta\Omega_{IE}^{L}$. Since $\Omega_{IE}^{L} = C_{E}^{L}\Omega_{IE}^{E}$ and $\delta\Omega_{IE}^{E} = 0$, we derive:

$$\delta\Omega_{IE}^{L} = \delta C_{E}^{L}\Omega_{IE}^{E}.$$

- But $\delta C_{E}^{L} = -E^{L}C_{E}^{L}$. Therefore,

$$\begin{aligned}\delta\Omega_{IE}^{L} &= -E^{L}C_{E}^{L}\Omega_{IE}^{E} \\ &= -E^{L}\Omega_{IE}^{L}.\end{aligned}$$

- For the error in transport rate, $\delta\Omega^L_{EL}$, using the error operator on Eq. (14), a new error terms having to do with altitude error shows up:

$$\begin{aligned}\delta\omega^L_{EL} &= \delta\left[\frac{1}{R}u^L_r \times v^L\right] \\ &= \frac{1}{R}u^L_r \times \delta v^L - \frac{\delta h}{R^2}\left[u^L_r \times v^L\right]\end{aligned}$$

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- Here, $R = R_e + h$ for a spherical earth of radius R_e.

Inertial Error Dynamics

Inertial Error Dynamics

- Eq. (24) can now be rewritten as:

$$\dot{\Phi}^L = -\left(\Omega^L_{IE} + \Omega^L_{EL}\right)\Phi^L - C^L_B \delta\Omega^B_{IB} + \Omega^L_{IE} E^L + \delta\Omega^L_{EL} \quad (27)$$

where $\delta\Omega^L_{EL}$ is the skew-symmetric matrix form of

$$\delta\omega^L_{EL} = \frac{1}{R}\left(u^L_R \times \delta v^L\right) - \frac{\delta h}{R^2}\left(u^L_R \times v^L\right)$$

- Similarly, the velocity equation can be rewritten:

$$\delta\dot{v}^L = C_B^L \delta a_{sf}^B + a_{sf}^B \times \phi^L - \left(2\omega_{IE}^L + \omega_{EL}^L\right) \times \delta v^L + \\ - \left(\delta\omega_{EL}^L - 2e^L \times \omega_{IE}^L\right) \times v^L + \frac{2g\delta h}{R_e} u_R^L + \delta g^L$$

while the equation for the position angle error e^L itself is

$$\dot{e}^L = -\Omega_{EL}^L e^L + \delta\omega_{EL}^L \tag{28}$$

- Further Simplifications Using Telescope Pointing Angle Error

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- $\psi^L = \phi^L - e^L$

- Further Simplifications Using Telescope Pointing Angle Error
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- Leads to:

- Further Simplifications Using Telescope Pointing Angle Error
- $\psi^L = \phi^L - e^L$
- Leads to:

$$\delta\dot{R}^L = \delta v^L - \Omega^L_{EL}\delta R^L \tag{29}$$

and

$$\delta\dot{v}^L = C^L_B\delta a^B_{sf} + a^B_{sf} \times \psi^L - \frac{g}{R_e}\Delta R^L_H - \left(2\omega^L_{IE} + \omega^L_{EL}\right) \times \delta v^L + \\ - \left(\delta\omega^L_{EL} - 2e^L \times \omega^L_{IE}\right) \times v^L + \frac{2g\delta h}{R_e}u^L_R + \delta g^L \tag{30}$$

- Finally, the equations for the telescope pointing angle errors are:

- Finally, the equations for the telescope pointing angle errors are:

$$\dot{\psi}^L = -\left(\Omega^L_{IE} + \Omega^L_{EL}\right)\psi^L - C^L_B \delta\omega^B_{IB} \tag{31}$$

- Three error equations yield three vector quantities of interest

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  - position error vector (3) δR derived from δC_L^E

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  - position error vector (3) δR derived from δC_L^E
  - velocity error vector (3) δV^L (or δv^L)
  - attitude error vector (3) ϕ_x, ϕ_y, ψ_z from δC_B^L

- See Matlab / Simulink Models

Inertial Error Dynamics

Strapdown Error Equations

- See Matlab / Simulink Models
  - Strapdown Navigation Equations: TBD

Inertial Error Dynamics

Strapdown Error Equations

- See Matlab / Simulink Models
  - Strapdown Navigation Equations: TBD
  - Strapdown Navigation Error Equations: TBD

Initialization: Alignment

- Coarse Leveling
  - Gimbal system limited by platform torquing rates
  - Strapdown systems have no real rate limits
  - Time average of accel outputs scaled by $1/g$
  - Yields third row of DCM to begin fine alignment
- Fine Alignment
- Alignment filter aids in further leveling

The Basic 9-State Kalman Filter

Topics

The Basic 9-State Kalman Filter

Topics

- Whole State Filters vs. Error State Filters

The Basic 9-State Kalman Filter

Topics

- Whole State Filters vs. Error State Filters
- The Canonical Nine States

The Basic 9-State Kalman Filter

Topics

- Whole State Filters vs. Error State Filters
- The Canonical Nine States
- INS Error State Dynamics and Transition Matrix

9-state Filter

Whole State vs. Error State Filters

9-state Filter

Whole State vs. Error State Filters

- Whole state filters ...

9-state Filter

Whole State vs. Error State Filters

- Whole state filters ...
  - model whole position, velocity, attitude

9-state Filter

Whole State vs. Error State Filters

- Whole state filters ...
  - model whole position, velocity, attitude
  - deal with large and rapidly varying quantities

9-state Filter

Whole State vs. Error State Filters

- Whole state filters ...
  - model whole position, velocity, attitude
  - deal with large and rapidly varying quantities
  - are highly non-linear

9-state Filter

Whole State vs. Error State Filters

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- Error state filters ...
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  - exhibit slowly varying dynamics: easier to linearize
  - have smaller quantities that allow us to take advantage of small angle assmptions

The Canonical Nine States

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- 3 position errors: $\delta R_x, \delta R_y, \delta R_z$ in local level coordinates

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- 3 attitude errors: ϕ_x, ϕ_y, ψ_z
  - ϕ_x, ϕ_y known as tilt errors
  - ψ_z known as azimuth error

Exercises

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- 1. Rewrite equation (24)

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L$$

to show the terms in the canonical order with position error, velocity error, and then attitude error.

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  to show the terms in the canonical order with position error, velocity error, and then attitude error.

- 2. Determine the 7th row of the system matrix F for the canonical 9-state error model. Use the cross-product operator matrix as required, e.g.,

$$\Omega_{IE}^L = \left[\, \omega_{IE}^L \times \,\right] = \begin{bmatrix} 0 & \left[\omega_{IE}^L\right]_z & -\left[\omega_{IE}^L\right]_y \\ -\left[\omega_{IE}^L\right]_z & 0 & \left[\omega_{IE}^L\right]_x \\ \left[\omega_{IE}^L\right]_y & -\left[\omega_{IE}^L\right]_x & 0 \end{bmatrix}.$$

  Introduce simplifying notation! E.g., set $\alpha = \omega_{IE}^L$ Hint: Use telescope pointint angle error equation (31)

Challenges of INS Filters

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- They don't help estimate aiding states such as Doppler bias and boresight errors
- Or GPS receiver clock errors
- They also don't help us to estimate inertial instrument drifts such as bias drift or scale factor errors

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- These are functions of time
- Position dependent items such as latitude are also involved

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- If Jacobian is evaluated at a nominal value of the trajectory, we call this a linearized filter
- If Jacobian is evaluated at the previous optimal estimate, we call it an Extended Kalman filter

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- And also choosing the right measurement scheduling
- Also, certain kinds of divergence can be the result of poor observability (more on this later)

Exercises

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- Use Eq. (29)

$$\delta \dot{R}^L = \delta v^L - \Omega^L_{EL} \delta R^L,$$

write the first three rows of the system dynamics matrix F in the continuous Kalman formulation:

$$\dot{x} = Fx + w$$